Take-Home Assignment

Product Manager

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About the Candidate

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Assignment Tasks

1. Google Analytics

From a user experience point of view, we want to evaluate Offers Page
This product is accessible mainly through the Gift Icon, but not limited through that. There are various entry points where you land on the Offers Homepage.
This product serves as a centralized hub for offers, helping players easily discover and navigate all available promotions — effectively acting as a one-stop shop for rewards and incentives.
Google Analytics can give us a hint or two with regards to Offers Page performance.

● Define what kind of events, kpis or funnels we can monitor, so we can extract valuable insights for the Product Team.
● Provide some reports, for illustrative reasons. (maybe using some random numbers)
● What success looks like for this product? Can you explain?
● Are there additional KPIs that, combined with your suggestions above,enhance our decision making
strategy?

Please check the following:
The relevant google sheet file (Offers Page for THA) with two tabs; Screenshots Desktop and Apps in
an illustrative way depicted how the player can browse through Offers Page and eventually choose a
game or an offer.
For extra tips, you can visit the Novibet site, based on your residency.
(for BR, https://www.novibet.bet.br/cassino)

1.1 Product Overview (key findings and analytical framing)

Since the market was regulated, Brazil has imposed a number of restrictions on the use of bonuses and promotions in the iGaming sector. This limits the creation of more appealing incentives and CTAs. The analysis will therefore take into account both the screenshots provided and an overview of the local website, which is better suited to legal and compliance requirements.

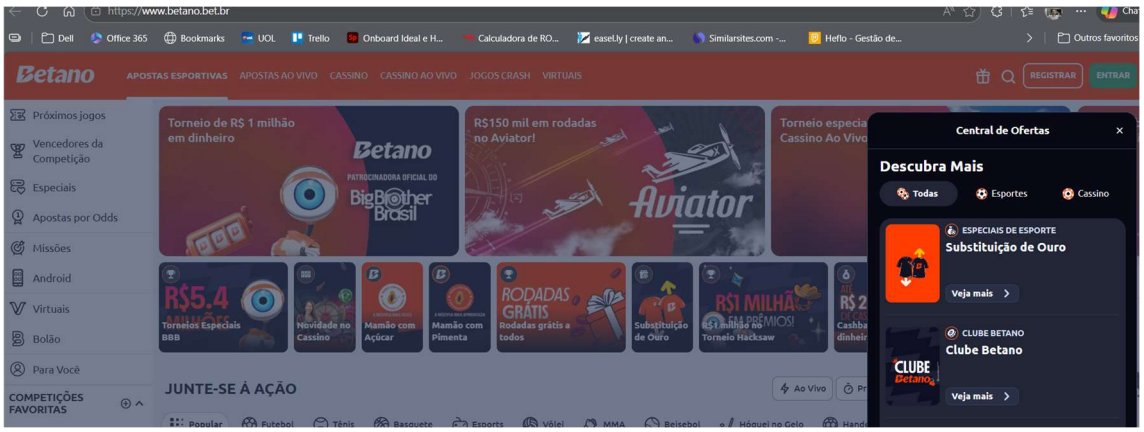
In both cases (screenshots and the Brazilian website), users access the Offer Page primarily via a Gift Icon in the navigation bar (both desktop and mobile app), but also through the Sportsbook Home Page and direct inner page links. This multi-entry architecture means the measurement framework must capture both traffic source quality and player journey depth from first discovery through to downstream betting activity.

The page serves two distinct player intents: discovery (players browsing available promotions without a specific offer in mind); and intent (players returning to claim a specific offer they have already seen). These intents require different event tracking and should inform separate funnel analyses.

The UX/UI structure is straightforward, based on an easily recognisable icon (the gift box). Offers can then be categorised (all, sports or casinos), which helps to segment the audience. In terms of structure, the main issues are the lack of a clear hierarchy (larger offers do not always appear at the top of the list) and the counters, which in some cases do not generate FOMO (the screenshot below shows offers with 496 and 1,747 days until expiry, which makes the measurement meaningless).

The brand's visual identity is reflected in the banners, and the colour scheme helps to make the page more appealing. I noticed the absence of CTAs or clickable buttons, such

as those found on Betano's Promotions Page (a market leader and benchmark in terms of bonuses and offers, which has a similar structure to the example provided, as can be seen below).



1.2 What to measure and monitor (KPIs, events, funnels)

Based on the proposed structure, the key KPIs to be monitored are:

- Website to Offer Page CVR (% of the total audience, broken down by desktop, mobile and app, who reach the Offer Page)
- Offer banner Click Rate
- Offer CTA Click Rate
- Offer to Register CVR
- Register to Deposit CVR
- Deposit to Bet CVR
- Exit Rate (churn / no action)
- Offer Expiration Rate
- Player performance (First deposit, GGR, NGR)
- Player LTV (average LTV x LTV from the Offer Page)

Generally speaking, the main concerns regarding the Offer Page should be:

- Whether it attracts the attention of visitors already on the site
- Whether visitors who land on it click on the offers
- Whether those who click on the offers actually play
- Whether these players represent good value when compared to the company's database
- Whether these high-value players have a good LTV

Regarding Funnel Exploration, I would create a 5-step report on GA with stages sequential and closed (each action requires the completion of the prior action). The stages should be:

	Step (event)	CVR	Drop-off signal (hypotheses)
1.	Offer Page visit (offer_page_view)		
2.	Offer banner click (offer_offer_click)	~40%	Low rate (copy, layout, content)
3.	CTA (new resource) engaged (offer_cta_primary_click)	~25%	Low rate (copy, content)
4.	Register completed (offer_opt_in)	~10%	Low rate (KYC, form page, tech)
5.	Downstream bet placed (bet_placed_post_offer)	~20%	Low rate (offer-bet mismatch, tech)

Figures based on personal experience with performance-based operations / market averages (information gathered from discussions with other professionals in the sector)

Based on this, a dashboard for monitoring this property might look like this:

Website to Offer	Offer banner CR	CTA CR
~8% *Page view → offer page*	**~2%** *Offer page → banner clicked*	**~5%** *Offer page → CTA clicked*

Offer to Register CVR	Register to Deposit CVR	Deposit do Bet CVR
~20% *Offer click → register*	**~70%** *Register → deposit*	**~50%** *Deposit → successful bet*

Exit Rate	Offer Expiration Rate
~60% *Offer Page sessions without CTA interaction*	**<25%** *Opted-in offers unused at expiry*

First deposit (AV)	GGR (AV)	NGR (AV)	LTV (AV)
>50 BRL *Based on Novibet's average results from performance campaigns*	**>60 BRL** *Based on Novibet's average results from performance campaigns*	**~60% GGR** *Based on Novibet's average results from performance campaigns*	**>60D** *Based on Novibet's average results from performance campaigns*

Figures based on personal experience with performance-based operations / market averages (information gathered from discussions with other professionals in the sector)

It is also important to follow the different results from each platform (desktop, mobile and app). The desktop experience displays significantly more offer detail per card (full descriptions, visual badges, clear CTAs) compared to the mobile app. This structural difference is expected to drive a measurable performance gap across devices. With that in mind, as the figures are fictitious, I have created a dashboard using AI (Claude) for reference and illustration purposes only:

Metric	Mobile App	Mobile Web	Desktop
Est. Share of Sessions	~50%	~28%	~22%
Avg. Scroll Depth	~45%	~52%	~71%
Offer Click Rate	~35%	~38%	~50%
Opt-in Rate (register)	~13%	~15%	~22%
Exit Rate (no action)	~66%	~63%	~50%

1.3 What "success" looks like in this case

Before regulation, the Brazilian iGaming market was built around bonuses and promotions, which proved to offer the best customer acquisition cost (CAC) in the sector. As there were no restrictions in place, many operators offered aggressive terms to attract new players, creating an environment where player retention was difficult (low lifetime value, LTV) and the average deposit amount was low (GGR dependent on the value of promotional offers).

The upside of this trend is that Brazilian players are very receptive to promotions. Therefore, we can expect positive conversion rates from the offers page.

On the other hand, it is important to note that players coming from offers pages are of lesser value to the operator than those from other sources. These are casual players who will subsequently require further retention efforts to keep them playing.

Taking this information into account, along with data collected in recent years, the product's success in this case could be assessed as follows:

Short-Term (Weekly)

- Offer Click Rate: 10% of page sessions across all devices
- Opt-in Rate: 25% of clicks
- Exit Rate (no action): below 60%

Medium-Term (Monthly)

- Offer-to-Bet Conversion Rate: 10% (all traffic sources combined)
- Redemption-driven GGR: 12% of total platform GGR
- CRM-driven sessions converting at ≥ 1.5× the organic entry rate (CRM traffic is higher-intent by definition)

Long-Term (Quarterly)

- Players engaging at least 2 distinct offers per month
- Offer Expiry Rate below 25% (currently untracked; baseline required)
- Bonus Cost Ratio (promotion spend / GGR attributed to offers): ≤ 25%

1.4 Additional KPIs for enhanced decision-making

- **Bonus cost ratio (BCR):** Controls profitability
- **Offer awareness rate:** Measures discoverability beyond gift icon
- **Multi-offer engagement:** Shows product stickiness

- **GGR per offer:** Enables comparison between the offers
- **CAC via offers:** Acquisition efficiency

2. Data Analysis & Product Insights

Now, from a more product point of view, we want to evaluate how Sportsbook Missions, one of our most engaging mechanics, is performing.
This product aims to motivate players to place a bet and they will get rewarded regardless if they win or lose.
We bring 13-weeks data from Brazil.
● Create some reports where you can demonstrate your analytical thinking.
● How are Missions performing? Is there any pain point spotted from the data?
● Provide additional KPIs that can be explored to enhance our decision making strategy.
● What success looks like? Can you explain?
● Can you provide any product improvement based on your data?
● If you were responsible to consult the product team for next steps, what would you say?
How would you respond if you were asked to drop the Missions product?
Please check the following:

The relevant google sheet file (Sportsbook Missions for THA) with 4 tabs. Definitions, Conversion, Customer Base, Category
For extra tips, you can visit the Novibet site, based on your residency.
(for BR, https://www.novibet.bet.br/apostas-esportivas)

2.1 Data overview (built with AI)

The dataset covers 13 weeks of Sportsbook Missions activity in Brazil (6 April 2025 – 29 June 2025), spanning three full calendar months. Four data tabs were provided: Definitions, Conversion, Customer Base, and Category. All figures in this section are drawn directly from the provided data; no illustrative substitution has been applied.

Numbers at a glance:

Total Opt-ins	Total Completions	Overall Conv. Rate	Total Missions Offered
181,540 *13 weeks, all missions*	**64,823** *13 weeks, all missions*	**35.7%** *Completions / opt-ins*	**99** *78 public · 21 private*

Active Users/Week (AV)	Opt-in Rate (AV)	User Completion Rate	Missions/Week (AV)
26,423 *Logged-in players*	**16.2%** *Opt-in users / active users*	**46.0%** *Completing users / opt-in users*	**7.6** *Range: 5–14 per week*

Source: Sportsbook Missions for THA — Conversion and Customer Base tabs.

Weekly Performance:

Week	Missions	Pub/Priv	Opt-ins	Completions	Conv. Rate	Active Users	Opt-in Rate
Apr 06	7	6/1	12,100	4,321	35.7%	23,000	15.9%
Apr 13	8	7/1	13,400	4,810	35.9%	23,120	16.0%
Apr 20	8	7/1	12,500	4,610	36.9%	24,100	15.1%
Apr 27	5	5/0	11,340	3,610	31.8%	25,320	15.4%
May 04	10	7/3	15,220	6,540	43.0%	25,800	15.9%
May 11	9	6/3	13,988	5,621	40.2%	27,100	15.9%
May 18	14	9/5	25,432	9,765	38.4%	29,100	15.3%
May 25	10	6/4	15,220	6,921	45.5%	28,200	16.3%
Jun 01	5	5/0	11,612	3,100	26.7%	27,830	16.7%
Jun 08	5	5/0	11,819	3,283	27.8%	27,820	16.5%
Jun 15	6	5/1	12,765	3,923	30.7%	27,340	16.9%
Jun 22	6	5/1	12,910	4,100	31.8%	27,360	17.0%
Jun 29	6	5/1	13,234	4,219	31.9%	27,410	17.3%

Source: Sportsbook Missions for THA — Conversion and Customer Base tabs. Conv. Rate = Completions / Opt-ins.

Monthly summary:

Month	Weeks	Missions	Opt-ins	Completions	Conv. Rate	Active Users (AV)

April	4	28	49,340	17,351	35.2%	23,885
May	4	43	69,860	28,847	41.3%	27,550
June	5	28	62,340	18,625	29.9%	27,552
TOTAL / AV	13	99	181,540	64,823	35.7%	26,423

2.2 What I gathered from the data

The most significant finding in the dataset is the sharp deterioration in the completion rate (opt-in conversion) during June: from a May average of 41.8% to a June average of 29.8%, a 12.0% retraction.

As it went beyond the margin, the fluctuation in the opt-in conversion rate suggests a structural breakdown. Particularly as there was no player disengagement. The opt-in rate (opt-in users / active users) was actually rising throughout June, from 16.7% to 17.3%. Players continued to engage with Missions at increasing rates. They simply stopped completing them.

So, looking at it more closely, May's peak performance (41.3% conv. rate) coincided with maximum mission diversity. Basketball accounted for 30% of missions and Mixed accounted for 33%. May 18, the single best week in the dataset (25,432 opt-ins; 9,765 completions), had 14 missions of which only 3 (21%) were Football, with 5 Basketball and 6 Mixed.

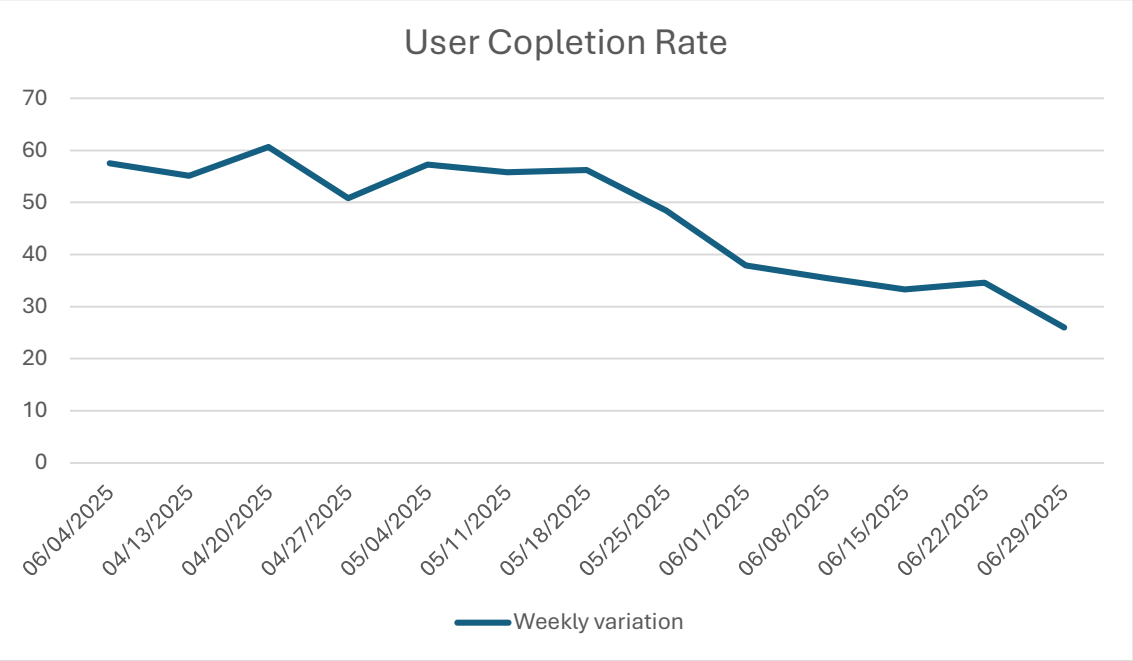
June's collapse maps precisely to the elimination of Basketball (0 missions) and Mixed (0 missions), leaving Football and Tennis as the only categories. The implication is clear: players who opt into multi-sport or Basketball missions complete them at materially higher rates. Football-only periods suppress completion, likely because football match schedules are less frequent and outcomes harder to control for within a 7-day mission window.

Month	Football Missions	Basketball Missions	Mixed Missions	CVR (AV)
April	16 (57%)	9 (32%)	0 (0%)	35.2%
May	15 (35%)	13 (30%)	14 (33%)	41.3%
June	13 (46%)	0 (0%)	0 (0%)	29.9%

The hypothesis that the chosen sports are a key factor in user behaviour and conversion rates is further supported by a specific week in April. During the period designated as 27 April, the missions were also limited to football and tennis. The number of completions fell from 4,610 to 3,610 WOW, and the conversion rate dropped from 36.9% to 31.8%.

On the other hand, it is important to note that football-related missions account for only 44.4% of the total analysed for the period. Although the sport accounted for the largest share (Basketball, in second place, accounted for half the missions), the percentage is small in terms of efficiency. Although the most reliable estimates are outdated, it is safe to say that football accounts for more than 80% of the sports betting market in Brazil. There is therefore a discrepancy between the market potential and the volume of Football Missions.

The completion rate also shows a worrying decline. The User Completion Rate (completing users / opt-in users) measures how many of the players who opted in actually completed at least one mission. The decline from 60.7% in April to 26% in the final week of June represents a 31% deterioration in user-level mission efficacy. This trend is not explained by category mix alone. It began during May, before the June category shift. It suggests a secondary issue, such as a possible mission fatigue or the level calibration drift (missions are becoming harder relative to player behaviour over time).



In addition, a correlation can be observed between the number of missions and the total number of opt-ins. On 27 April and 1 June, when there were only five missions, the number of opt-ins stood at around 11,000 (11,340 and 11,612, respectively). In May, on the other hand, there were 14 missions and 25,432 opt-ins. Although this affects the average number of opt-ins per mission, the figures show that the ideal is to have around 10–12 missions per week.

And the final point to note is that weeks featuring only public missions performed less well in terms of conversion (32.4% vs 40.4% for weeks with private missions).

In summary, therefore, it was possible to observe that:

- There was a drop in the completion rate from June onwards
- The sports chosen affect the completion rate (greater diversity = higher completion rate)
- Conversely, football is underutilised (44.4%, far below the share the sport represents in the Brazilian sports betting market)
- User Completion Rate dropped from 67% in April to 26% in May (Fewer people are completing missions, which suggests there is a problem with the missions themselves)
- More missions = more sign-ups. The figures show that, for a few weeks, the number of missions was lower than ideal (above 10)

- All-public missions weeks underperformed on conversion (Weeks with 0 private missions average 32.4% CR vs 40.4% for weeks with private missions)

2.3 Prioritisation of the findings

Completion rate decline	**CRITICAL**
Category concentration	**CRITICAL**
User completion rate decline	**HIGH**
Football being underutilised	**HIGH**
Mission volume below optimal	**MEDIUM**
All-public missions weeks underperform on conversion	**MEDIUM**

2.4 What "success" looks like in this case

In terms of the product, the data points to a clear path to success:

- Private sessions available in at least 75% of weeks
- Football featuring more prominently in the total number of sessions
- Greater variety of sports among the sessions on offer

The main aim of the proposed changes is to halt the decline in the completion rate and the completion rate per user. Furthermore, the aim is to return to the peak levels of opt-ins (25,432) and users (29,100) recorded on 18 May.

2.5 Additional KPIs or decision-making

The figures presented highlight other KPIs that can be monitored to gauge the actual impact of the campaigns on the brand strategy as a whole.

It is possible, for example, to measure the average GGR per opt-in and the overall performance of players acquired through missions (primarily NGR and LTV).

Another key aspect is retention. What percentage of people return organically for subsequent missions? What is the recirculation rate?

It is also possible to look at strategic breakdowns (Which game type converts best? What is the actual impact of private missions on the conversion rate?).

Finally, it is important to understand that missions are merely a gateway. So, what percentage of people leave this platform and continue betting? What is the performance of these players?

2.6 Executive assessment

The product generated 64,823 completions in 13 weeks from a base of 26,000 weekly active users. The opt-in rate of 16.2% is structurally stable even in the worst weeks. The

critical insight from this data is that Missions does not have a product problem, but a content curation problem. June's collapse was caused by removing Basketball and Mixed missions, not by any fundamental product flaw. The correct recommendation is to fix the category mix, not drop the product. A controlled A/B experiment (Missions On vs. Off for matched cohorts over 6 weeks) would provide the causal evidence needed for any go/no-go decision.

This analysis must also take into account the impact that gamification has on the profile of the website's visitors. The mission format helps to make the entire site more appealing to a younger audience, who are likely to develop a longer-lasting and more profitable relationship with the brand.

Looking at the results, I would invest in the following:

- Content adjustments (game modes, volume of missions, increased focus on football, increased focus on private missions)
- Product analysis (ongoing research into opt-in and conversion rates)
- Analysis of additional KPIs (assessment of the potential of players recruited through missions)
- Inclusion of other commercial opportunities (CTAs to other parts of the site, upsell opportunities and bonuses/cashback linked to mission results)

Given that the campaigns attract more than 26,000 people a week and that this figure shows no signs of waning, it seems clear to me that this is a platform with the potential to drive repeat engagement and link up with the company's other commercial opportunities.

3. How to?

In Novibet, we launched a Casino Loyalty program in January 2025.
It is a way of rewarding our players, whenever they launch and play a game.
There is no requirement for the player (like opt-in) - he/she instantly receives coins based on their activity.
The player starts with 0 coins and there is no competition among the players.
These coins can be automatically redeemed based on the below tables.
The "Premium" is a one-off stage; the player receives rewards as soon as a milestone is reached.
The "VIP" is a monthly recurring stage; the player receives rewards by the end of month, based on his/her highest milestone achieved during each month.
The player will complete the "Premium" stage, before moving to the "VIP" stage.

(there is no need to focus so much on the numerics)

The Product Analytics team has been requested to assess if the Casino Loyalty program is successful or not, until now (End of August 2025).
You are responsible to compile a concrete analysis depicting the as is performance of this casino loyalty.

- **How would you frame the analysis?**
- **Explain the perimeter of your analysis e.g. time period, customer segment, etc**
- **Feel free to provide any visuals, if necessary.**

For extra tips, you can visit the Novibet site, based on your residency.
(for BR, https://www.novibet.bet.br/cassino/loyalty-program)

Novibet Casino Loyalty Programme, launched January 2025, rewards players automatically for game activity (no opt-in required). Players accumulate coins from launch, progressing through a one-off Premium stage (5 milestones) before entering the monthly-recurring VIP stage (7 milestones). Rewards are cash-denominated and auto-redeemed at each milestone. The assessment scope covers January 2025 through end of August 2025 (8 months).

Note: *No operational data for this product was provided in the assignment. The framework below describes how the analysis should be structured and what data would be required.*

3.1 Conceptual analysis

This is a less practical aspect of the assessment, but it is important for establishing the product's foundation and its significance beyond tangible factors.

Brazil has 28 million active players, with ~37 billion BRL in GGR (source: SIGAP / SPA). Of this total, 68.3% are men and 51.3% are aged between 18 and 40. The South-East region accounts for almost half of all players (São Paulo alone accounts for 23.4%).

Therefore, given that the industry's primary target audience is men aged between 18 and 40 living in the South-East (primarily in São Paulo), the Casino Loyalty Programme should take the following into account:

- Gamification matters: This audience profile is closely linked to gaming culture. According to the Game Brasil Survey, conducted in 2022, 74.5% of the Brazilian population plays video games. The demographic most actively engaged in gaming is men aged between 16 and 34.
- Dopamine release builds loyalty: From another perspective, the rewards system plays a key role in audience retention strategies. Small rewards trigger dopamine release in the audience without straining the company's finances, and this constant release plays a significant role in the retention process (encouraging the player to keep playing).
- Customer loyalty reduces CAC: As betting operators sell commodities, one of the most significant costs of any operation lies in customer acquisition (promotion, PR, storytelling and brand awareness, relationship building, funnel structuring, and guiding the audience towards conversion). Therefore, any initiative with the potential to reduce CAC consequently has a major impact on operational efficiency.
- Loyalty opens up new business opportunities: Once players are engaged with the platform and continue to seek out the next rewards, a range of commercial opportunities becomes available. For example: you can charge for shortcuts, experiences or features, as games do, or create scenarios where players can compete against others (with rewards based on performance and/or referrals).
- Loyalty and clustering are trends: One of the concepts most frequently discussed in recent digital business debates is that of 'dark forests'. Essentially, people have been relating to the internet as if they were alone in a dark forest: everything feels threatening, the chances of encountering a predator or bot are high, and the opportunities for genuine connections are slim. This has been one of the driving forces behind the rise of closed communities (Discord, Telegram, WhatsApp) at the expense of posts on open platforms. It is also one of the justifications for business deals such as the New York Times' acquisition of The Athletic, which paid over US$550 million in 2022 to gain access to a subscriber base of 1.2 billion. Thus, the business value of such initiatives also lies in the number of loyal users.

This introduction is necessary to make the point that, regardless of the figures involved, loyalty programmes are extremely important for strategic reasons. This understanding must form the starting point for any analysis of a specific scheme.

3.2 Analytical framework

Having established the conceptual foundations for the programme's justification, an analytical framework should have four priorities:

- Activation: Are players discovering and progressing into the programme? How much does it cost to promote the programme and attract new players?
- Engagement: Are players advancing through milestones meaningfully? What is the average number of stages each of them has completed?
- Revenue generation: Does loyalty drive net-new GGR beyond what the player would have generated? How does the GGR of loyalty scheme members compare with that of other players? How does it compare when looking solely at casino players?
- Retention: Does milestone progression reduce churn? How does the average monthly deposit of players in the programme compare with that of other players? What is the average stake placed by players in the programme and by other players?

Finally, it is essential to take into account operating costs (the number of people involved, hours allocated, and tools hired) and the total revenue generated by the programme.

3.3 Perimeter of the analysis

The period under review spans eight months, but requires an important caveat. The programme was launched in January, the first month of the regulated market in Brazil. Due to the need for adaptation, this was a period of significant negative impact across all market metrics. The Blask Index, for example, fell 28% in January 2025.

Therefore, even though the analysis covers an eight-month period, it is important to make it clear that January's result is likely to be an outlier.

For the purposes of analysis, I propose segmenting the audience into four profiles:

- New players – People who made their first deposit between January and August 2025
- Established players – People who were already active on the platform before January 2025
- Reactivated players – People who had no activity on the platform in the 60 days prior to the launch of the programme
- Control group – Matched cohort from pre-programme period (e.g., Aug–Dec 2024)

3.4 Key issues raised in the areas analysed

- Activation: Because enrolment is passive (no opt-in required), 100% of casino players are technically enrolled. The meaningful activation question is therefore: what share of players actually earn enough coins to reach Premium Milestone 1 (2,000 coins)?

- Engagement: The Premium stage structure (milestones at 2,000 / 7,000 / 17,000 / 42,000 / 92,000 coins) contains a structural risk: the gap between M3 and M4 is 25,000 coins, which is 2.5x the prior interval (M2/M3 = 10,000 coins). This non-linearity is expected to cause a pronounced drop-off at M4.
- Revenue generation: This is the most critical dimension for justifying programme cost. The analysis must control for self-selection bias: players who earn more coins are inherently more active. A naive comparison (loyalty players vs. non-loyalty players) will overstate the programme's causal effect.
- Retention: Each Progressive milestone should correlate with measurably lower churn. The key hypothesis is that players who have completed a milestone are more "anchored" to the platform due to sunk-cost investment in coin accumulation and anticipation of the next reward.

3.5 Key proposals to tackle the issues raised

- Activation: Operational and promotional costs (what is the cost of bringing a player into the programme, and what is the cost of getting them to accumulate their first 2,000 coins?)
- Engagement: Plot milestone completion funnel by monthly cohort. If M3/M4 drop-off exceeds M2/M3 drop-off by more than 2x, introduce an interim milestone at ~28,000/30,000 coins to reduce perceived distance.
- Revenue generation: Compare the GGR trajectory of players who reached each milestone vs. a matched cohort of players with similar pre-programme activity who did not reach that milestone (control group). This controls for the fact that high-activity players self-select into higher milestones.
- Retention: Compare the average GGR generated by players in the programme (from M2 onwards) with the average GGR of the control group. Compare the NGR of players in the programme with the overall NGR. Compare the LTV of players in the programme with the overall LTV.

3.6 Risks to follow

- Acquisition cost
- Drop-off after the each milestone
- Retention cost (communication, promotion, monitoring)
- Reactivation cost (communication, promotion, monitoring)
- Programme cost (is it better than other onboarding platforms, such as bonuses?)

4. North Star Metric Framework

The way we evaluate the performance of our products is through the NSM framework.
NSM stands for North Star Metric; we can have multiple metrics to evaluate a product, but you need to choose one metric that best reflects whether the product is truly growing and delivering value.

- **Propose a NSM for one of the above mentioned products (Offers Page, Sportsbook Missions, Casino Loyalty)**
- **Explain your rationale and for choosing this metric**
- **Identify 3-4 additional, secondary metrics that can either influence the NSM or cover another dimension of success for the product**
- **Describe the way you can actually use this NSM framework to guide product decisions and prioritise initiatives.**

4.1 Rationale for the chosen metric

The three case studies presented in this document cover two main subjects. The first (Offers Page) deals with user acquisition, whilst the other two (Sportsbook Missions and Casino Loyalty) focus more on customer retention. The difference is that one of them (Casino Loyalty) uses rewards to capture the audience's attention, whilst the other (Sportsbook Missions) is based on the repetition of tasks or challenges.

Therefore, my first criterion for selection was to identify the type of product in which I have the greatest confidence. I truly believe that customer loyalty will be a major driver of efficiency in iGaming over the coming years. This is a sector where there is little product differentiation, and the relationship between brands and customers is essentially transactional (offers, reactivation, bonuses, complaints, issues, rewards). The Brazilian market has a huge volume of players, but the average GGR is low compared to countries where the industry is more mature. Furthermore, as I pointed out above, the Brazilian iGaming market has been built around companies that acquire users at extremely high costs. So, any initiative that builds player loyalty and reduces CAC has a massive impact on operational efficiency.

With this in mind, I can use my experience and knowledge of the market as a preliminary filter for the figures.

So, looking more closely at the logic behind Sportsbook Missions, it becomes clear that the metrics can also be grouped. Some capture only intent, such as clicks and opt-ins, but do not necessarily indicate a business impact. Others, such as the number of missions undertaken or the number of missions completed, depend on decisions made by parties other than the players (if the platform decides to offer more missions, they will be able to hit the target regardless of the conversion rate). Finally, some targets are far removed from business impact (audience, number of active players, % of missions completed).

With this in mind, the metric I have chosen to apply within the NSM methodology is **weekly transacting mission users (WTMU)**, which represents the number of unique players who entered one or more missions (opt in) and completed one or more missions within the specified timeframe (one week). This, of course, takes into account only transacting missions (missions with a business impact).

It is important to note that this is a complex metric, unlike those presented in the financial reports and dashboards, and is designed to provide an in-depth insight into the operation in question.

Furthermore, A player who opts in and completes a Mission has engaged with the full product cycle. They experienced the incentive, placed qualifying bets, and received a reward. This is not a passive impression.

WTMU captures the players most likely to generate incremental GGR and retain long-term. The mission completion event is the point at which the product has demonstrably influenced player behaviour.

4.2 Secondary Metrics (input metrics)

These four metrics directly influence WTMU. Each one covers a dimension WTMU alone cannot capture:

- Opt-in Conversion Rate (Completions / Opt-ins): Covers mission difficulty calibration
- User Completion Rate (Completing users / Opt-in users): Follows the individual player efficiency
- Category Diversity Index (1 - HHI* of mission type share): Measures portfolio risk + market alignment
- Private Mission Share (Private missions / Total missions): Understands the personalisation depth

** HHI = Herfindahl-Hirschman Index, a standard measure of category concentration. A score of 1.0 indicates complete concentration in one category; 0 indicates perfect distribution. Used here as a simple diversity proxy.*

4.3 Using the NSN Framework to guide decisions

I asked the AI (Claude) to draw up a four-step plan for WTMU to follow in the specified project. The steps I included in the prompt were:

Step 1 — Weekly WTMU as the diagnostic trigger
Step 2 — Prioritise initiatives by expected WTMU impact
Step 3 — Quarterly OKR structure against WTMU
Step 4 — A/B Test Governance

The answers he gave are as follows:

Step 1 — Weekly WTMU as the diagnostic trigger
If WTMU declines week-over-week, consult the four secondary metrics to identify the cause before making any product change.

If this input metric declines…	The diagnosis is…	The intervention is…
Opt-in Conversion Rate	Mission difficulty too high or category mismatch	Add Basketball/Mixed missions; review odds thresholds
User Completion Rate	Fatigue, UX friction, or mission design drift	Survey non-completers; review mission progress UI
Category Diversity Index	Over-reliance on football calendar	Mandate minimum non-football missions per week
Private Mission Share drops to 0	Segmentation abandoned in planning	Reinstate private missions for ≥2 slots per week

Step 2 — Prioritise initiatives by expected WTMU impact
When competing for product development resources, rank initiatives by their estimated WTMU impact, validated by the June data:

- Mandate category diversity (Basketball/Mixed every week) → est. +30–40% WTMU recovery to May levels, based on observed data.
- Halt User Completion Rate decline (investigate fatigue/difficulty) → est. +10–15% WTMU if rate stabilised at April levels.

- Reinstate private missions every week → est. +8 pp completion rate uplift, translating to ~+5% WTMU.
- Increase weekly mission volume to 9–12 → est. +10–20% absolute WTMU via additional completion opportunities.

Step 3 — Quarterly OKR structure against WTMU

Quarter	WTMU Target	Primary Lever	Owner
Q2 2025 (baseline)	Establish measurement; target: ~2,000/week	Instrument WTMU tracking; validate data pipeline	Analytics + Product
Q3 2025	+25% vs Q2 avg	Category diversity mandate; private missions reinstated	Missions Planning + Product
Q4 2025	+15% vs Q3 avg	Mission difficulty recalibration; User Completion Rate recovery	Product + UX
Q1 2026	+10% vs Q4 avg	Always-on rolling mission structure; personalisation layer	Product + CRM

Step 4 — A/B Test Governance

Every product change to Missions should be validated as a randomised controlled experiment with WTMU as the primary outcome metric. Minimum requirements: 95% confidence threshold; minimum detectable effect of 5% relative lift; minimum 2-week test duration (4 weeks for low-volume segments). Secondary metrics serve as guardrails, an initiative that improves WTMU but degrades User Completion Rate or BCR beyond thresholds should not be shipped.

4.4 Executive assessment

Taken together, the initial exercise carried out, the metric identified and the framework created by AI demonstrate that it is possible to build customer loyalty by focusing on delivery, thereby reducing CAC and increasing player LTV. This would also open up the possibility for the company to adopt an even more youthful, dynamic and gamified approach to communication, creating a direct link with younger audiences (the iGaming market is restricted to those aged 18 and over).

In this regard, it is particularly important to align the company's OKRs with the project's outcomes to ensure the desired impact during the months of implementation.